

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)	$n = \frac{pV}{RT} = \frac{1.2 \times 10^5 \times 1.6 \times 10^{-3}}{8.31 \times 293.0} = [0.0789 \text{ mol}] \text{ (1)}$ $nN_A = 0.0789 \times 6.02 \times 10^{23} = 4.75 \times 10^{22} \text{ (1)}$ Alternative: $N = \frac{pV}{kT} = \frac{1.2 \times 10^5 \times 1.6 \times 10^{-3}}{1.38 \times 10^{-23} \times 293.0} \text{ (1)}$ $= 4.75 \times 10^{22} \text{ (1)}$		2		2	2	
		(ii)	Valid method e.g. $\frac{1}{2}mv^2 = \frac{3}{2}kT \text{ (1)}$ Answer = 1 350 [m s ⁻¹] ecf on $N \text{ (1)}$	1	1		2	2	
	(b)	(i)	Combine: $nRT = \frac{1}{3}Nm\overline{c^2} \text{ (1)}$ Substitution of $N = nN_A$ or $n = 1$ so $N = N_A \text{ (1)}$ KE [of a molecule] = $\frac{1}{2}m\overline{c^2}$ or $U = N\frac{1}{2}m\overline{c^2}$ can be implied if everything is clear (1) So use or implied $k = \frac{R}{N_A}$ and convincing algebra (1) Common alternative for full marks: $nRT = \frac{1}{3}Nm\overline{c^2} \text{ (1)}$ $NkT = \frac{1}{3}Nm\overline{c^2} \text{ (1) (implies 1st line if 1st line omitted)}$ $KE = \frac{1}{2}m\overline{c^2} \text{ (1) (can be implied if everything clear)}$ Algebra leading to $KE = \frac{3}{2}kT \text{ (1)}$	4			4	2	